DO NOT WRITE IN THIS AREA

10 Peter pays £5000 for a factory machine.

The value of the machine depreciates at a rate of x% per annum.
At the end of 6 years the value of the machine is £3493.62

Calculate the value of x.
Give your answer correct to 1 decimal place.

Let the yearly multiplier be k.

$5000k^6 = 3493.62$

1 mark
Divide by the starting value.

$k^6 = 3493.62 \div 5000 = 0.698724$

$k = \sqrt[6]{(0.698724)} = 0.942000088...$

1 mark
Take the sixth root.

$x = (1 - k) \times 100 = 5.799991177...$

$x = 5.8\%$ (1 decimal place)

1 mark
Convert decrease to %.

(Total for Question 10 is 3 marks)

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